MILITARY SPECIFICATION

INDICATOR, TEMPERATURE, THERMOCOUPLE, ENGINE EXHAUST, TYPE MJ-1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers a single-pointer thermocouple-type temperature indicator, designated type MJ-1.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation. Methods of.
MIL-I-7071	Indicator, Temperature, Thermocouple, Hermetically Sealed, General Specification for.

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of.
MS33639	Cases, Instrument, Clamp Mounted, Aircraft.
MS35215	Screw, Machine-Pan Head, Cross-Recessed, Brass, Black Oxide, UNF-2A.
MS35338	Washer, Lock-Spring, Helical, Regular (Medium) Series.
MS51957	Screw, Machine-Pan Head, Cross-Recessed Corrosion Resisting Steel, Passivated, UNC-2A.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

AIR FORCE-NAVY AERONAUTICAL STANDARD DRAWINGS

AN961	Washer - Flat, Electrical

PUBLICATIONS

AIR FORCE-NAVY AERONAUTICAL BULLETIN

No. 405	Storage Life - Aeronautical Articles

* (Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.1.3 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951	Standard Commercial Packaging

* (Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

* 2.1.4 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. The requirements specified in specification MIL-I-7071 are applicable as requirements of this specification. Additional requirements shall be as specified herein.

3.2 Qualification. The indicator furnished under this specification shall be a product which has been tested and has passed the qualification tests specified.

3.3 Design. The indicator shall conform to figure 1.

3.4 Construction.

3.4.1 Case back. The back of the indicator case shall be distinctly marked so as to indicate "plus" for the chromel-terminal stud and "minus" for the alumel-terminal stud. The statement, "Use 22-ohm special leads," shall be plainly marked on the rear of the case.

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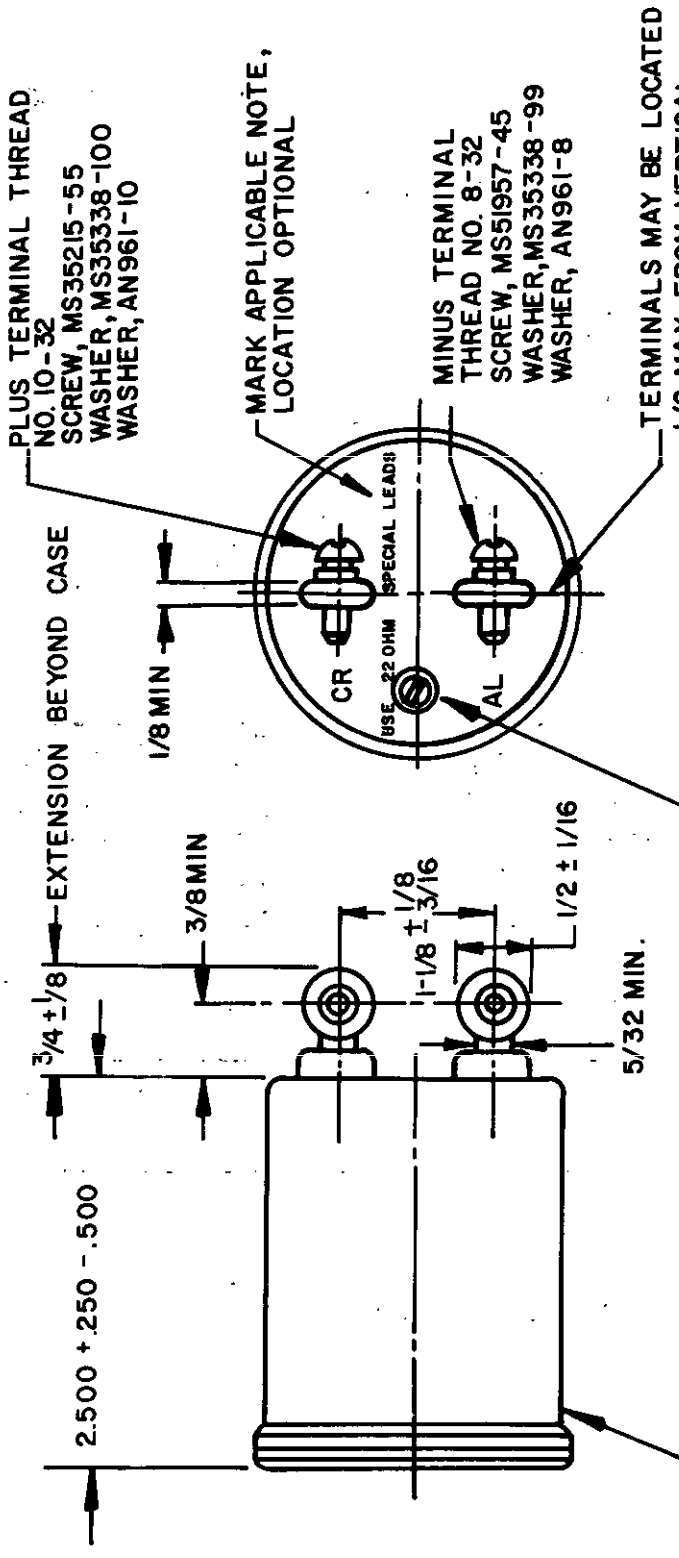
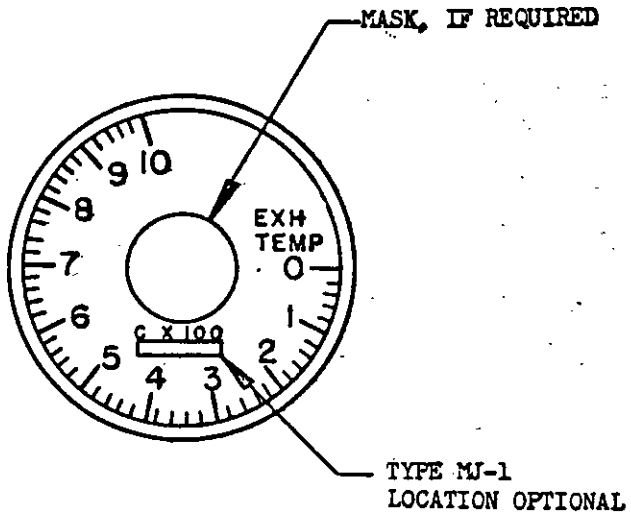


FIGURE 1. INDICATOR DETAILS

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MARKINGS	HEIGHT OR LENGTH ±0.005	WIDTH OF LINE OR GRADUATION ±0.005	MATERIAL OR FINISH
NUMBERS	0.156	0.020	
ALL MAJOR GRADUATIONS	0.156	0.020	
ALL MINOR GRADUATIONS	0.093	0.015	FLUORESCENT LUMINESCENT MATERIAL
LETTERING			
"EXH TEMP"	0.125	0.020	
"C X 100"	0.093	0.015	
BACKGROUND OF DIAL TYPE MJ-1	0.062	0.010	DURABLE DULL BLACK

DIMENSIONS IN INCHES.

FIGURE 2. Dial details

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TABLE I

Exhaust temperature test tolerances

Test point	Scale error at room temperature	Scale error at low temperature	Scale error at high temperature	Scale error at room temperature after vibration failure	Extreme low temperature exposure		
					Scale error at low temperature	Scale error at room temperature after low temperature	Scale error at room temperature after high temperature
°C	Tolerance ±°C	Tolerance ±°C	Tolerance ±°C	Tolerance ±°C	Tolerance ±°C	Tolerance ±°C	Tolerance ±°C
0	40 (40)	50 (50)	50 (50)	50 (50)	60 (50)	50 (40)	50 (40)
200	30 (30)	40 (40)	40 (40)	40 (40)	50 (40)	40 (35)	40 (35)
400	20 (20)	35 (35)	30 (30)	35 (30)	45 (40)	40 (30)	40 (30)
500	20 (20)	35 (35)	30 (30)	35 (30)	45 (40)	40 (30)	40 (30)
600	10 (10)	25 (25)	20 (20)	25 (20)	40 (30)	30 (20)	30 (20)
700	5 (5)	20 (20)	15 (15)	20 (15)	35 (25)	25 (15)	25 (15)
800	10 (10)	25 (25)	20 (20)	25 (20)	40 (30)	30 (20)	30 (20)
900	20 (20)	35 (35)	30 (30)	35 (30)	45 (40)	40 (35)	40 (25)
1,000	30 (30)	40 (40)	40 (40)	40 (35)	45 (40)	40 (35)	40 (35)

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3.4.2 Dial. The dial of the indicator shall conform to figure 2.

3.4.3 Scale arc. The scale arc of the indicator shall be not less than 250 degrees. The mechanism shall not be geared to obtain the specified scale arc.

3.4.3.1 Scale distribution. The scale of the indicator shall be as linear as is practicable and shall not be constricted or expanded by more than 1 angular degree between any two adjacent 20°C scale graduations within the range of 200° to 900°C. The remainder of the scale shall not be constricted or expanded by more than 1.5 angular degrees between any 2 adjacent scale graduations. The scales are based on a minimum of 5 angular degrees nominal spacing between 20°C scale graduations.

3.4.4 Pointer. The width of the pointer 1/2 ±1/16 inch from the tip shall be 5/64 ±1/64 inch and shall taper to a point. The pointer shall be finished with fluorescent-luminescent material.

3.4.4.1 Pointer overswing. The pointer overswing shall be not less than 5 angular degrees deflection beyond each end of the scale.

3.5 Weight. The weight of the complete indicator shall not exceed 1.0 pound.

3.6 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General. Sampling, inspection, and testing of the indicator shall be as specified in specification MIL-I-7071, and as specified herein.

* 4.1.2 Qualification tests. Qualification testing shall consist of all the tests as specified in general specification (4.6 of MIL-I-7071) and as specified herein (see 4.3).

4.2 Test tolerances. The only tolerances specified in table I, not enclosed in parentheses, are applicable only when the indicators are subjected to all the tests in the sequence specified herein. The tolerances specified in table I, enclosed in parentheses, are applicable only when the indicators are subjected to a specific test independent of sequence.

4.3 Acceptance tests. Acceptance testing shall consist of the individual tests (4.3.1), sampling tests, (4.3.2) and if specified by the procuring activity, the special tests (4.3.3) of this specification.

4.3.1 Individual tests.

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4.3.1.1 Scale error at room temperature. The scale errors at the test points specified in table I shall not exceed the tolerances specified in table I. When this test is conducted, the pointer shall be set to read 700°C when the millivoltage for 700°C, as specified in table I of specification MIL-I-7071 for the ambient room temperature, is applied to the terminals of the indicator through the appropriate resistance in series with the indicator. If more accurate over-all indicator accuracy can be obtained by setting the pointer to 700° ±5°C with the specified voltage for 700°C, the pointer shall be so positioned.

4.3.1.2 Position error. This test may be conducted at any point on the scale between 200° and 900°C by applying test voltage. The maximum change in indication shall not exceed 25°C. This test shall be conducted at a major or minor scale graduation.

4.3.1.3 Friction. The maximum difference in reading, before and after tapping, shall not exceed 30°C.

4.3.1.4 Zero adjustment. Power shall be applied to produce a 700°C reading. When the indicator zero adjustment is operated throughout its range, the resulting pointer motion shall be not less than 40°C above and below the 700°C reading. The total motion of the pointer shall not exceed 200°C. The pointer shall be reset to the initial reading with the initial power applied.

4.3.1.5 Insulation leakage. The insulation resistance shall be a minimum of 20 megohms.

4.3.2 Sampling tests.

4.3.2.1 Scale error at low temperature. The scale errors shall not exceed the tolerance specified in table I. If there is any tendency for the pointer to stick to the pointer stops, a light tap on the indicator shall be sufficient to free the pointer.

4.3.2.2 Scale error at high temperature. The scale errors shall not exceed the tolerances specified in table I. If there is any tendency for the pointer to stick to the pointer stops, a light tap on the indicator shall be sufficient to free the pointer.

4.3.2.3 Magnetic effect. When the indicator is subjected to the magnetic effect test, it shall not cause the magnetic compass to deflect more than 5 degrees.

4.3.2.4 Vibration error. The maximum amplitude of pointer oscillation throughout the frequency range shall not exceed a range of 25°C within the temperature range of 400° to 900°C and shall not exceed 30°C for the remainder of the scale. The maximum difference between the original position of the pointer before vibration and the position of the pointer during vibration shall not exceed 15°C between 400° and 900°C and shall not exceed 25°C for the remainder of the scale. Pointer oscillation shall not be considered as pointer variation. The diameter of the circular motion of vibration shall be 0.009 to 0.011 inch.

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* 4.3.2.5 Vibration failure. The scale errors at room temperature, after vibration, shall not exceed the tolerance specified in table I. The zero adjuster shall not be reset after this vibration test.

4.3.2.6 Resistance. The resistance of the indicator shall be 33 + 5, -0 ohms.

4.3.3 Special tests. Unless specified by the procuring activity, those sources listed on the applicable Qualified Products List will not be required to conduct these tests.

4.3.3.1 Cold junction compensator lag. The indicator shall be connected to a chromel-alumel thermocouple (22 ohms total external-circuit resistance), immersed in an ice bath of 0°C, and a millivoltage of constant value applied in series with the indicator and thermocouple to provide readings of 700°C. The indicator shall then be subjected to the conditions specified in specification MIL-I-7071 for increasing and decreasing temperatures. The indication of the indicator throughout the test shall at no time vary by more than 30°C from the indication obtained at room temperature. When conducting this test, copper-constantan leads shall be connected to the terminals of the indicator.

4.3.3.2 Extreme low temperature exposure. The scale errors at low temperature shall not exceed the tolerances specified in table I. The scale errors at room temperature, after low temperature exposure, shall not exceed the tolerance specified in table I.

4.3.3.3 High temperature exposure. The scale errors at room temperature after high temperature exposure shall not exceed the tolerances specified in table I.

5. PACKAGING.

* 5.1 Preservation. Preservation shall be level A, C, or commercial as specified (see 6.3).

* 5.1.1 Level A. Each indicator shall be preserved and packaged method IC in accordance with MIL-P-116.

* 5.1.2 Level C. The level C preservation for indicators shall conform to the requirements of MIL-STD-794 for this level.

* 5.1.3 Commercial. The commercial/industrial preservation of indicators shall be in accordance with the requirements of ASTM D 3951.

* 5.2 Packing. Packing shall be level A, B, C or commercial, as specified (see 6.3).

* 5.2.1 Level A. The packaged indicators shall be packed in containers as specified in MIL-STD-794 for level A protection.

* 5.2.2 Level B. The packaged indicators shall be packed in containers as specified in MIL-STD-794 for level B protection.

* 5.2.3 Level C. The packaged indicators shall be packed in containers as specified in MIL-STD-794 for level C protection.

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* 5.2.4 Commercial. The packaged indicators shall be packed in accordance with the requirements of ASTM D 3951.

5.3 Marking.

5.3.1 Levels A, B and C. In addition to any special or other identification marking required by the contract (see 6.3), each unit pack, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

* 5.3.2 Commercial. Commercial/industrial marking shall be in accordance with the requirements of ASTM D 3951.

6. NOTES

6.1 Intended use. The type MJ-1 indicator covered by this specification is intended for use in indication exhaust temperature of jet engines.

6.1.1 The temperature indicator can be used on instrument panels requiring case mounting, in accordance with drawing AND10403, by the application of an instrument mounting bezel in accordance with specification MIL-B-6836.

* 6.2 Ordering data.

* 6.2.1 Data Requirements. Acquisition documents should specify the following: Applicable levels of preservation, packaging and packing (see paragraph 5.1 and 5.2).

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List (QPL No.) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is SA-ALC/MMEDO, Kelly AFB, TX 78241, and information pertaining to qualification of products may be obtained from that activity.

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

* 6.5 Certain provisions of this specification are the subject of international standardization Air Standard 10/63. When amendment, revision or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate

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action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-9343C	Indicator, Temperature Thermocouple, Engine, Exhaust, Type MJ-1

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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